

Project name: supply chain management

Date: 1/10/2025 10:47:56 AM

The Art and Science Steps of Songleading Fuzzy DEMATEL Method [Specialty Track 1]

Step 1: Generate the fuzzy direct relation matrix

In order to identify the model of the relations among the n criteria, an $n \times n$ matrix is first generated. The influence of the element in each row exerted on the element in each column of this matrix can be represented a fuzzy number. If multiple experts' opinions are used, all experts must complete the matrix. arithmetic mean of all of the experts' opinions is used to generate the direct relation matrix Z .

$$Z = \begin{bmatrix} 0 & \dots & \tilde{z}_{1n} \\ \vdots & \ddots & \vdots \\ \tilde{z}_{n1} & \dots & 0 \end{bmatrix}$$

The table below indicates the direct relation matrix, which is the same as pairwise comparison matrix of the experts.

The direct relation matrix

<u>AUTHOR(S):</u>	Noam Katz	Quality	Cost	Delivery Reliability	Sustainability
<u>SUMMARY:</u> In this intensive, we will learn the Art (and Science) of Songleading through three focused areas: Song Leading skills, Repertoire and Creating Community through Music. – Submitted by Noam Katz (No instrument or musical ability necessary!) (0.000,0.000,0.000)			(0.250,0.500,0.750)	(0.000,0.250,0.500)	(0.000,0.000,0.250)

CORNERSTONE 2017 RESOURCE

<u>TO</u> <u>PIC</u> <u>(S):</u> Cost	<u>Music & Rhythm:</u> <u>Community</u> <u>Building</u> (0.000,0.250,0.500)	(0.000,0.000,0.000)	(0.250,0.500,0.750)	(0.000,0.250,0.500)
<u>LE</u> <u>AR</u> <u>NIN</u> <u>G</u> <u>OBJ</u> <u>EC</u> <u>TIV</u> <u>E:</u> Deliver try Relia bility	Participants will gain knowledge in how to effectively teach new music, build a song session, use an array of verbal and non-verbal cues, and gain new camp repertoire(0.250,0.500,0.750)	(0.000,0.250,0.500)	(0.000,0.000,0.000)	(0.250,0.500,0.750)
<u>AU</u> <u>DIE</u> <u>NC</u> <u>E:</u> Sustain ability	35-40 participants(0.000,0.000,0.250)	(0.250,0.500,0.750)	(0.250,0.500,0.750)	(0.000,0.000,0.000)
<u>TIM</u> <u>ING</u> :	90			
<u>APP</u> <u>EN</u> <u>DIC</u> <u>ES:</u>	<u>Cornerstone 2017 Songleading Track (Appendix #1); CS Chord Companion (Appendix #2); How NOT to Teach a Song (Appendix #3)</u>			
<u>MA</u> <u>TE</u> <u>RIA</u> <u>LS</u> <u>NE</u> <u>ED</u> <u>ED:</u>	<u>Shoebox with lid and mirror inside, Butcher paper, markers, guitar, music stand, drums/percussion</u>			
<u>SET</u> <u>-UP</u> <u>DET</u>	<u>Indoor space large enough for circle of chairs, and small group work (including space for instruments)</u>			

SESSION TIMELINE & OUTLINE:

- | | |
|--|------------|
| 1. Intentional Drum Circle | (15 mins.) |
| 2. Introductions – impactful musical moments at camp | (5 mins.) |
| 3. Why songlead? Ben Folds interview | (10 mins.) |
| 4. What's in Your Guitar Case? | (10 mins.) |
| 5. Song Teaching: Do's & Don'ts | (25 mins.) |
| 6. Repertoire | (15 mins.) |
| 7. Mirror in a Box/Drumming | (10 mins.) |

1. Intentional Drum Circle (15 mins.)

Participants walk in with drums/percussion already in place on chairs. Play/drum together to build sense of community through music and rhythm. Facilitator gives cues when to start, stop, play louder/softer. If time permits, play “Call-and-Response” or “Funky in the Middle” games to engage.

- Why do you think we began a Songleading track with a drum circle? What might these activities have in common? (music/rhythm, build community with shared sense of purpose, FUN! relaxing, establishes the roles of facilitator and group, etc.)

2. Introductions (5 mins.)

Go around the circle. Everyone introduces their names, camps, Songleading experience (if any) and ONE impactful musical moment in their lives. *Doesn't have to be Jewish or at camp.*

3. Why Songlead? Ben Folds interview (10 mins.)

- What role does music play at camp?
- What are the benefits of having a skilled songleader (as opposed to spontaneous group singing)? Important for ruach at camp, education, holy moments/worship, transmitters of Hebrew text, facilitator to bring out the best in the group, etc.
- Why do YOU want to do it?

Listen to Ben Folds' NPR interview from 2001.

- What attitudes/techniques does he use to transform his solo “living room” concerts into opportunities for group singing?
- What are the differences between performance and songleading? Similarities?

4. What's in Your Guitar Case? (10 mins.)

Facilitator asks, “What are the basic skills and values needed to be an effective songleader?”

Have someone draw a large guitar case on piece of butcher paper. Then everyone takes markers writes words or draws symbols to answer the question.

Facilitator adds: *Da lifnei mi atah omeid lifnei atah omeid. Know before whom you stand...before you*

CORNERSTONE 2017 RESOURCE

If time permits, do index card activity in chevruta (as outlined on second-to-last page of Appendix #1)

5. Song Teaching: Do's & Don'ts (25 mins.)

Distribute slips from Appendix #3: "How NOT to Teach a Song".

Play until most participants have had a turn.

Teach a song (from Appendix #2).

- What worked well for you in learning that song?
- What could have been improved?

Review elements of good song teaching on pages 1-3 of Appendix #1. Include mention of:

- Preparing words and chords *ahead* of time
- Knowing the meaning or context (especially if it's in Hebrew!)
- Introduce with enthusiasm (the "sell")
- Start with the hook/chorus
- Teach melody without the words (or, occasionally, rhythm of words without melody)
- RAM: "Repeat after me" / Chunking
- Review, review, review (lines, sections, tricky parts)
- Troubleshooting
- 24-hour rule: "the Stickiness Factor"

If time, ask for any volunteers to teach a short song (or small part of one).

6. Repertoire (15 mins.)

Teach new camp repertoire from Appendix #2 and/or slides.

7. Mirror in a Box / Drumming (10 mins.)

Pass around a shoebox with a mirror hidden inside (participants do not know it is there). Ask each participant to look inside and say 2 things that are very visible in the box and 1 thing that is inside, but much more difficult to see. (e.g., "I can easily see confidence and a sense of humor, but it is more difficult to see humility.") Because this is a very introspective exercise, it is encouraged that the facilitator(s) go first to set a tone. Some will be too uncomfortable to take it seriously (e.g., "I see two ears and a nose"), but don't scold or get frustrated. Use it as a learning opportunity in your follow-up questions:

- Were you surprised by what was inside the shoebox?
- What was most difficult about this activity for you? Why?
- What do you think was the point of this exercise? What does it have to do with songleading?

If time remains, end with drumming to bookend the workshop.

CORNERSTONE 2017 RESOURCE

The following table shows the fuzzy scale used in the model.

Fuzzy Scale

Code	Linguistic terms	L	M	U
1	No influence	0	0	0.25
2	Very low influence	0	0.25	0.5
3	Low influence	0.25	0.5	0.75
4	High influence	0.5	0.75	1
5	Very high influence	0.75	1	1

Step 2: Normalize the fuzzy direct relation matrix

The normalized fuzzy direct relation matrix can be obtained using the following formula:

$$\tilde{x}_{ij} = \frac{\tilde{z}_{ij}}{r} = \left(\frac{l_{ij}}{r}, \frac{m_{ij}}{r}, \frac{u_{ij}}{r} \right)$$

where

$$r = \max_{i,j} \left\{ \max_i \sum_{j=1}^n u_{ij}, \max_j \sum_{i=1}^n u_{ij} \right\} \quad i, j \in \{1, 2, 3, \dots, n\}$$

The normalized fuzzy direct relation matrix

	Quality	Cost	Delivery Reliability	Sustainability
Quality	{0.000,0.000,0.000}	{0.125,0.250,0.375}	{0.000,0.125,0.250}	{0.000,0.000,0.125}
Cost	{0.000,0.125,0.250}	{0.000,0.000,0.000}	{0.125,0.250,0.375}	{0.000,0.125,0.250}
Delivery Reliability	{0.125,0.250,0.375}	{0.000,0.125,0.250}	{0.000,0.000,0.000}	{0.125,0.250,0.375}
Sustainability	{0.000,0.000,0.125}	{0.125,0.250,0.375}	{0.125,0.250,0.375}	{0.000,0.000,0.000}

Step 3: Calculate the fuzzy total relation matrix

In step 3, the fuzzy total relation matrix can be calculated by the following formula:

$$\tilde{T} = \lim_{k \rightarrow +\infty} (\tilde{x}^1 \oplus \tilde{x}^2 \oplus \dots \oplus \tilde{x}^k)$$

If each element of the fuzzy total relation matrix is expressed as $\tilde{t}_{ij} = (l_{ij}^{\tilde{t}}, m_{ij}^{\tilde{t}}, u_{ij}^{\tilde{t}})$, it can be calculated as follows:

$$[l_{ij}^{\tilde{t}}] = x_l \times (1 - x_l)^{-1}$$

$$[m_{ij}^{\tilde{t}}] = x_m \times (1 - x_m)^{-1}$$

$$[u_{ij}^{\tilde{t}}] = x_u \times (1 - x_u)^{-1}$$

In other words, the normalized matrix the inverse is first calculated, and then it is subtracted from the matrix I, and finally the normalized matrix is multiplied by the resulting matrix. The following table shows the fuzzy direct relation matrix.

The fuzzy total relation matrix

	Quality	Cost	Delivery Reliability	Sustainability

Quality	(0.002,0.103,1.337)	(0.125,0.332,1.877)	(0.016,0.247,1.831)	(0.002,0.103,1.448)
Cost	(0.016,0.242,1.724)	(0.004,0.169,1.828)	(0.127,0.383,2.138)	(0.016,0.242,1.724)
Delivery-Reliability	(0.127,0.343,1.931)	(0.032,0.322,2.207)	(0.020,0.209,2.034)	(0.127,0.343,1.931)
Sustainability	(0.018,0.146,1.663)	(0.129,0.373,2.123)	(0.143,0.398,2.169)	(0.018,0.146,1.552)

Step 4: Defuzzify into crisp values

The CFCS method proposed by Opricovic and Tzeng has been used to obtain a crisp value of total relation matrix. The steps of CFCS method are as follows:

$$l_{ij}^n = \frac{(l_{ij}^t - \min l_{ij}^t)}{\Delta_{min}^{max}}$$

$$m_{ij}^n = \frac{(m_{ij}^t - \min l_{ij}^t)}{\Delta_{min}^{max}}$$

$$u_{ij}^n = \frac{(u_{ij}^t - \min l_{ij}^t)}{\Delta_{min}^{max}}$$

So that

$$\Delta_{min}^{max} = \max u_{ij}^t - \min l_{ij}^t$$

Calculating the upper and lower bounds of normalized values:

$$l_{ij}^s = \frac{m_{ij}^n}{(1 + m_{ij}^n - l_{ij}^n)}$$

$$u_{ij}^s = \frac{u_{ij}^n}{(1 + u_{ij}^n - l_{ij}^n)}$$

The output of the CFCS algorithm is crisp values.

Calculating total normalized crisp values:

$$x_{ij} = \frac{[l_{ij}^s(1 - l_{ij}^s) + u_{ij}^s \times u_{ij}^s]}{[1 - l_{ij}^s + u_{ij}^s]}$$

The crisp total relation matrix

	Quality	Cost	Delivery-Reliability	Sustainability
Quality	0.319	0.598	0.517	0.338
Cost	0.492	0.456	0.672	0.492
Delivery-Reliability	0.605	0.631	0.519	0.605
Sustainability	0.408	0.664	0.689	0.391

Step 5: set the threshold value

The threshold value must be obtained in order to calculate the internal relations matrix. Accordingly, partial relations are neglected and the network relationship map (NRM) is plotted. Only relations whose values in matrix T is greater than the threshold value are depicted in the NRM. To compute the threshold value for relations, it is sufficient to calculate the average values

CORNERSTONE 2017 RESOURCE

of the matrix T. After the threshold intensity is determined, all values in matrix T which are smaller than the threshold value are set to zero, that is, the causal relation mentioned above is not considered.

In this study, the threshold value is equal to 0.5250.525

All the values in matrix T which are smaller than 0.5250.525 are set to zero, that is, the causal relation mentioned above is not considered. The model of significant relations is presented in the following table.

The crisp total relationships matrix by considering the threshold value

	Quality	Cost	Delivery Reliability	Sustainability
Quality	0	0.598	0	0
Cost	0	0	0.672	0
Delivery Reliability	0.605	0.631	0	0.605
Sustainability	0	0.664	0.689	0

Step 6: Final output and create a causal relation diagram

The next step is to find out the sum of each row and each column of T (in step 4). The sum of rows (D) and columns (R) can be calculated as follows:

$$D = \sum_{j=1}^n T_{ij}$$

$$R = \sum_{i=1}^n T_{ij}$$

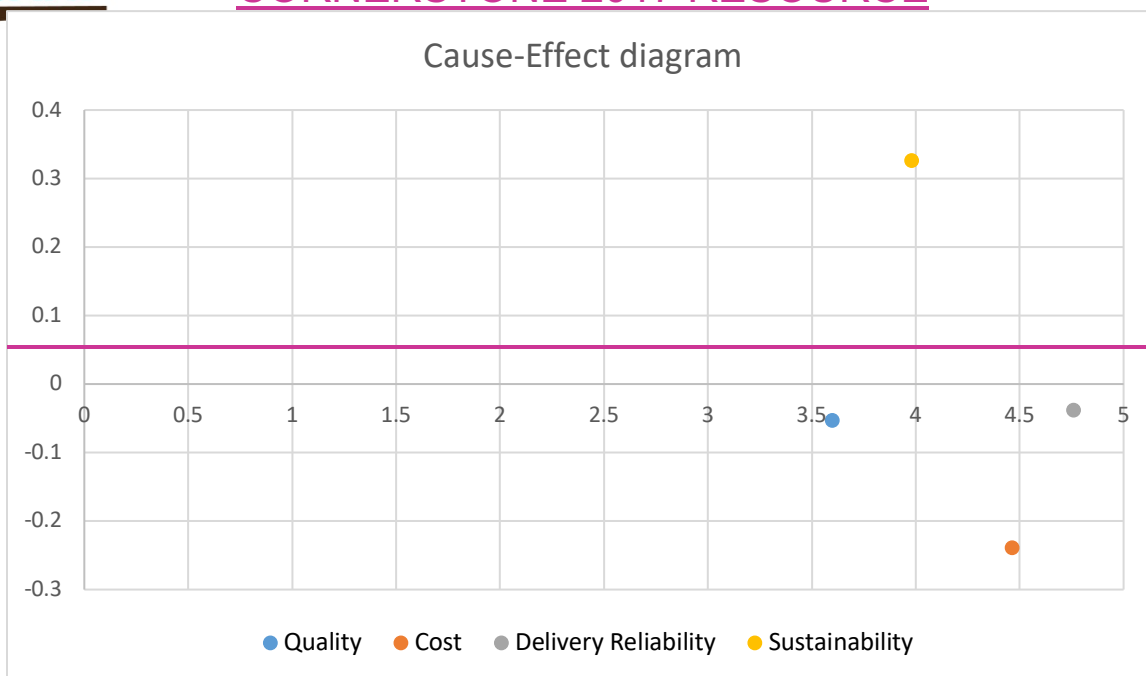
Then, the values of D+R and D-R can be calculated by D and R, where D+R represent the degree of importance of factor i in the entire system and D-R represent net effects that factor i contributes to the system.

The table below shows the final output.

The final output

	R	D	D+R	D-R
Quality	1.824	1.772	3.597	-0.052
Cost	2.35	2.112	4.461	-0.238
Delivery Reliability	2.398	2.36	4.758	-0.037
Sustainability	1.826	2.153	3.978	0.327

The following figure shows the model of significant relations. This model can be represented as a diagram in which the values of (D+R) are placed on the horizontal axis and the values of (D-R) on the vertical axis. The position and interaction of each factor with a point in the coordinates (D+R, D-R) are determined by coordinate system.



cause-effect diagram

Step 7: Interpret the results

According to the diagram and table above, each factor can be assessed based on the following aspects:

—Horizontal vector ($D + R$) represents the degree of importance between each factor plays in the entire system. In other words, ($D + R$) indicates both factor i's impact on the whole system and other system factors' impact on the factor. in terms of degree of importance, Delivery Reliability is ranked in first place and Cost, Sustainability and Quality, are ranked in the next places. In this study, Sustainability are considered to be as a causal variable, Quality, Cost, Delivery Reliability are regarded as an effect.

—The vertical vector ($D - R$) represents the degree of a factor's influence on system. In general, the positive value of $D - R$ represents a causal variable, and the negative value of $D - R$ represents an effect. in terms of degree of importance, Delivery Reliability is ranked in first place and Cost, Sustainability and Quality, are ranked in the next places. In this study, Sustainability are considered to be as a causal variable, Quality, Cost, Delivery Reliability are regarded as an effect.